

Lesson 59 Practice: Current CPU Architecture, Data Representation & RISC/CISC

JKSSB Computer Awareness

October 10, 2026

Provenance: 17 original JKSSB-pattern questions and five paraphrased adaptations of the JKSSB Junior Assistant 2026 computer block (Q62, Q64, Q65, final key). The adaptations at Q1, Q5, Q7, Q11, and Q13 are reworded, not verbatim reproductions.

- Q1.** *[PYQ-adapted: JKSSB Junior Assistant 2026 Q62; final key; paraphrased]* Which statement describes von Neumann (stored-program) architecture?
- A. Instructions and data are stored in two separate memories.
 - B. Only data is stored in memory; instructions are not.
 - C. Instructions and data are stored together in the same memory.
 - D. Only instructions are stored in memory; data is not.
- Q2.** *[Original]* In immediate addressing, the operand is:
- A. held in a general-purpose register.
 - B. contained within the instruction itself.
 - C. stored in main memory at an address named by the instruction.
 - D. the address of the next instruction to fetch.
- Q3.** *[Original]* The cache-hit ratio is defined as:
- A. hits divided by total accesses.
 - B. misses divided by total accesses.
 - C. hits divided by misses.
 - D. total accesses divided by hits.
- Q4.** *[Original]* Why is a cache hit faster than a cache miss?
- A. The cache is located on secondary storage.
 - B. The cache stores only instructions, never data.
 - C. The cache is smaller and faster and sits closer to the CPU, so its access time is lower than main memory.
 - D. The cache runs at the same speed as a hard disk.
- Q5.** *[PYQ-adapted: JKSSB Junior Assistant 2026 Q62; final key; paraphrased]* Which statement about data hazards is correct?

- A. A hazard can arise only when an operand is in memory.
- B. Register addressing removes every possible hazard.
- C. A hazard can arise only with immediate addressing.
- D. A hazard can still arise from a dependency between instructions, for example in a pipeline.

Q6. [Original] Which option correctly contrasts cache and virtual memory?

- A. Cache is on secondary storage, while virtual memory is in the cache.
- B. Cache and virtual memory are the same piece of hardware.
- C. Virtual memory is a single cache line.
- D. Cache is small fast memory between the CPU and main memory; virtual memory makes memory appear larger by using secondary storage as backing store.

Q7. [PYQ-adapted: JKSSB Junior Assistant 2026 Q64; final key; paraphrased] The Program Counter holds:

- A. the address of the instruction currently being executed.
- B. the address of the next instruction to be fetched.
- C. the operand value of the current instruction.
- D. the last instruction that finished executing.

Q8. [Original] The Instruction Register holds:

- A. the address of the next instruction to fetch.
- B. the instruction currently being fetched, decoded, and executed.
- C. the value of the operand.
- D. the width of the data bus.

Q9. [Original] Interrupts are normally processed:

- A. only after the entire program has completed.
- B. only during system boot.
- C. only for hardware faults.
- D. at instruction boundaries, during program execution.

Q10. [Original] Which statement about instruction timing is correct?

- A. Every machine instruction takes exactly one clock cycle.
- B. The clock-cycle count is fixed for the whole program.
- C. Different machine instructions can take different numbers of clock cycles.
- D. Machine instructions do not consume clock cycles.

Q11. [PYQ-adapted: JKSSB Junior Assistant 2026 Q65; final key; paraphrased] In two's-complement arithmetic, overflow occurs when:

- A. there is a carry out of the most significant bit.
- B. the result of an addition is exactly zero.

- C. the operation is performed on unsigned numbers.
 D. two operands of the same sign produce a result of the opposite sign.
- Q12.** *[Original]* When widening a value to more bits, which rule is correct?
- A. Zero-extension is for signed values and sign-extension is for unsigned values.
 B. Zero-extension is for unsigned values and sign-extension is for signed values.
 C. Both unsigned and signed values use zero-extension.
 D. Both unsigned and signed values use sign-extension.
- Q13.** *[PYQ-adapted: JKSSB Junior Assistant 2026 Q65; final key; paraphrased]* In BCD (binary-coded decimal), the decimal number 25 is represented as:
- A. the plain binary value 11001_2 .
 B. eight bits per decimal digit.
 C. the hexadecimal value 25_{16} .
 D. 0010 0101, one 4-bit code for each decimal digit.
- Q14.** *[Original]* Which statement about byte-addressability and the effective address is correct?
- A. The instruction length is always equal to the effective address.
 B. Memory is word-addressable only, so individual bytes have no address.
 C. Each byte has its own address, and the effective address is the operand's actual location, which need not equal the instruction length.
 D. The effective address is always equal to the Program Counter value.
- Q15.** *[Original]* What are the full forms of RISC and CISC?
- A. RISC = Reduced Instruction Set Computer; CISC = Complex Instruction Set Computer
 B. RISC = Rapid Instruction Set Computer; CISC = Complex Instruction Set Computer
 C. RISC = Reduced Integer Storage Cache; CISC = Central Instruction Set Computer
 D. RISC = Random Instruction Set Computing; CISC = Common Instruction Set Computer
- Q16.** *[Original]* Which set of features describes a RISC processor?
- A. Many complex, variable-length instructions.
 B. Instructions that always operate directly on memory operands.
 C. Microprogrammed control of highly complex instructions.
 D. Simple, mostly fixed-length instructions, many registers, and a load/store architecture.
- Q17.** *[Original]* Which set of features describes a CISC processor?
- A. A few simple fixed-length instructions only.
 B. A large set of complex, variable-length instructions, often allowing memory operands.
 C. No general-purpose registers at all.
 D. A strict load/store-only memory access rule.

Q18. *[Original]* Which statement is **NOT** correct?

- A. The Program Counter holds the address of the next instruction to fetch.
- B. Interrupts can be handled at instruction boundaries.
- C. All machine instructions take the same number of clock cycles.
- D. Two's-complement overflow can occur when two same-sign operands are added.

Q19. *[Original]* Which code correctly represents the decimal digit 5 in BCD?

- A. 0101
- B. 1001
- C. 101
- D. 1110

Q20. *[Original]* Evaluate the statements:

- I. In immediate addressing, the operand is part of the instruction.
- II. The cache-hit ratio equals hits divided by total accesses.
- III. Two's-complement overflow is the same as a carry out of the most significant bit.

- A. I and II only
- B. II and III only
- C. I and III only
- D. I, II, and III

Q21. *[Original]* The von Neumann bottleneck refers to:

- A. the delay between keyboard input and screen output.
- B. the limited rate at which instructions and data can be transferred between the CPU and memory over a shared path.
- C. the time taken for a hard disk to spin up.
- D. the number of general-purpose registers in the CPU.

Q22. *[Original]* The main purpose of virtual memory is to:

- A. increase the number of CPU registers.
- B. replace the cache entirely.
- C. reduce the CPU clock speed.
- D. make the available memory appear larger than physical RAM by using secondary storage as backing store.

Q23. *[JKSSB FAA-pattern]* RISC designs are characterised by:

- A. a small set of simple instructions executed quickly.
- B. hundreds of complex printer commands.
- C. manual paper calculation only.

- D. slower clocks by definition.
- Q24.** *[JKPSI-pattern]* CISC designs are characterised by:
- A. many complex multi-step instructions.
 - B. no use of memory at all.
 - C. operations only on paper.
 - D. absence of any control unit.
- Q25.** *[SSC CGL-pattern]* Which processor family follows the x86 CISC tradition?
- A. Intel Core desktop processors
 - B. Paper calculators
 - C. Mechanical typewriters
 - D. Wall clocks
- Q26.** *[JKSSB-pattern]* (MSQ: select ALL correct options.) Which statements about RISC are correct?
- A. Instructions are simple and uniform.
 - B. Most execution uses registers (load-store design).
 - C. Pipelining is easy to implement well.
 - D. RISC forbids the use of any memory.
- Q27.** *[SSC CGL-pattern]* (MSQ: select ALL correct options.) Which statements comparing RISC and CISC are correct?
- A. RISC uses fewer, simpler instructions than CISC.
 - B. CISC instructions can complete complex tasks in fewer lines.
 - C. ARM designs follow RISC ideas.
 - D. CISC processors contain no control unit.
- Q28.** *[PYQ-adapted: SSC CGL 2025 Tier-II Paper-I, 19-Jan-2026 Q6; SRC-18]* Which register holds the instruction currently being decoded and executed?
- A. MAR
 - B. PC
 - C. IR
 - D. Accumulator
- Q29.** *[PYQ-adapted: JKSSB Junior Assistant 2026 Q65, final key; SRC-03/SRC-06]* In BCD, each decimal digit is represented using:
- A. Four bits
 - B. Eight bits
 - C. One bit
 - D. Sixteen bits

Q30. [PYQ-adapted: JKSSB Junior Assistant 2026 Q65, final key; SRC-03/SRC-06] For unsigned widening, which method fills the newly added high-order bits with zeros?

- A. Zero extension
- B. Sign extension
- C. Sign inversion
- D. Carry propagation

Q31. [Original] In a pipelined processor, a data hazard can occur when:

- A. An instruction depends on a result not yet produced by an earlier instruction
- B. A printer runs out of paper
- C. A file extension is hidden
- D. A keyboard key is repeated

Q32. [Original, MSQ drill] (MSQ: select ALL correct options.) Which are standard CPU architecture concepts?

- A. Program Counter stores the next instruction address.
- B. Instruction Register holds the current instruction.
- C. A carry-out and signed overflow are always identical.
- D. BCD uses four bits per decimal digit.